

PGI-RC Correction

What the correction does

The gap IORad – IOLib equals the share of total variance attributed to PGI EA. Because the PGI is a noisy proxy for the true additive genetic factor g^* , this share is attenuated. The correction (Becker et al. (2021)) rescales it upward using $\rho > 1$, the ratio of true genetic signal to observed PGI signal:

$$\rho = \sqrt{\frac{h_{\text{SNP}}^2}{R_{\text{PGI}}^2}} > 1$$

Because our decomposition distinguishes between-family and within-family contributions, we need two separate correction factors — ρ_{between} and ρ_{within} — each computed from the corresponding h^2 and R^2 at that level.

Step 1. External heritability estimates (h^2)

We use two scenarios, differing in the source of h_{pop}^2 . Both are for educational attainment measured in years of schooling.

Scenario A — Becker et al. (2021), WLS-specific

- $h_{\text{pop}}^2 = 0.172$ (SE = 0.103): WLS-specific GREML estimate, Supplement Table 4.
- h_{within}^2 : Becker does not report a within-family estimate. We impute it by scaling Howe’s within/total ratio to Becker’s larger h_{pop}^2 , on the assumption that the share of genetic variance transmitted within families is constant across populations:

$$h_{\text{within,Becker}}^2 = h_{\text{within,Howe}}^2 \times \frac{h_{\text{pop,Becker}}^2}{h_{\text{pop,Howe}}^2} = 0.04 \times \frac{0.172}{0.13} \approx 0.053$$

- $h_{\text{between}}^2 = h_{\text{pop}}^2 - h_{\text{within}}^2 \approx 0.119$

Modeling choice: the proportional scaling assumption is not testable from the available data. It produces a Becker scenario that is internally consistent (within/total ratio matches Howe), but the true within-family h^2 for WLS could differ. Using Howe’s $h_{\text{within}}^2 = 0.04$ directly would overestimate h_{between}^2 (since Becker’s total is larger), and was therefore not used.

Scenario B — Howe et al. (2022), Nature Genetics

The data sources included in the estimations include MoBa (see Supplementary Table 1).

- $h_{\text{pop}}^2 = 0.13$ (95% CI: 0.12–0.15)

- $h^2_{\text{within}} = 0.04$ (95% CI: 0.02–0.05)
- $h^2_{\text{between}} = h^2_{\text{pop}} - h^2_{\text{within}} = 0.09$

This scenario is internally consistent: both h^2 components come from the same study and population. It is the more conservative scenario (smaller h^2 , smaller correction).

Additive decomposition

h^2_{pop} decomposes additively because h^2_{between} and h^2_{within} are both fractions of total phenotypic variance — the numerators partition the total genetic variance with the same denominator throughout:

$$h^2_{\text{pop}} = h^2_{\text{between}} + h^2_{\text{within}}$$

Step 2. Estimating R^2 from the data

We need two R^2 estimates from the WLS sibling sample: one for the between-family association between PGI and education, and one for the within-family association.

Between-family R^2

1. Residualize both education and PGI on $\text{poly}(\text{birth_year}, 2) \times \text{sex}$ and the top 10 genetic principal components (PCs) (Becker et al. 2021).
2. Compute family means of both residuals.
3. Regress mean education residual on mean PGI residual using OLS. R^2_{between} is the R^2 of this regression.

Within-family R^2

1. Residualize both education and PGI on $\text{poly}(\text{birth_year}, 2) \times \text{sex}$ only. **No PCs** are included.
2. Demean both residuals within each family, isolating within-family variation.
3. Regress demeaned education on demeaned PGI using OLS. R^2_{within} is the R^2 of this regression.

ICC

We estimate the intraclass correlation (ICC) from an intercept-only mixed model, and use it in Step 3 to put h^2 and R^2 on the same scale.

Consistency check

Because R^2_{between} and R^2_{within} use different denominators (between- and within-family variance respectively), they combine as a weighted average to give the population-level R^2 :

$$R_{\text{pop}}^2 = \text{ICC} \times R_{\text{between}}^2 + (1 - \text{ICC}) \times R_{\text{within}}^2$$

This differs from the additive h^2 decomposition because R^2 components use level-specific denominators while h^2 components all use total variance. Becker et al. report $R_{\text{pop}}^2 = 0.063$; our estimate is $R_{\text{pop}}^2 = 0.065$.

Step 3. Computing ρ

h_{between}^2 and h_{within}^2 from the literature are fractions of *total* phenotypic variance. R_{between}^2 and R_{within}^2 from our data are fractions of *between-* and *within-family* variance respectively. We convert R^2 to total-variance scale before dividing:

$$\rho_{\text{between}} = \sqrt{\frac{h_{\text{between}}^2}{\text{ICC} \times R_{\text{between}}^2}}, \quad \rho_{\text{within}} = \sqrt{\frac{h_{\text{within}}^2}{(1 - \text{ICC}) \times R_{\text{within}}^2}}$$

Step 4. Applying the correction

See PGI_RC_derivations for the full derivation.

Notation

Symbol	Definition
$\sigma^2(\tau'_j)$	Between-family variance from M2 (conditioning on PGI + ascribed)
w	$[\sigma^2(\delta_{ij}^*) - \sigma^2(\delta'_{ij})]/\sigma_{\text{tot}}^2$ — within-family share explained by ascribed characteristics
Δ_{between}	$[\sigma^2(\tau_j) - \sigma^2(\tau_j^*)]/\sigma_{\text{tot}}^2$ — observed between-family genetic share
Δ_{within}	$[\sigma^2(\delta_{ij}) - \sigma^2(\delta_{ij}^*)]/\sigma_{\text{tot}}^2$ — observed within-family genetic share
ρ_{between}	$\sqrt{h_{\text{between}}^2/(\text{ICC} \times R_{\text{between}}^2)}$
ρ_{within}	$\sqrt{h_{\text{within}}^2/((1 - \text{ICC}) \times R_{\text{within}}^2)}$

Corrected IOLib

$$\text{IOLib}_{\text{RC}} = \text{IOLib} - (\rho_{\text{between}}^2 - 1) \Delta_{\text{between}}$$

Since $\rho_{\text{between}} > 1$, $\text{IOLib}_{\text{RC}} < \text{IOLib}$: the noisy PGI left too much variance in $\sigma^2(\tau_j^*)$, over-attributing shared-family-environment variance to the liberal measure.

Corrected IORad

$$\text{IORad}_{\text{RC}} = \text{Sibcorr} + \rho_{\text{within}}^2 \Delta_{\text{within}} + w$$

Since $\rho_{\text{within}} > 1$, $\text{IORad}_{\text{RC}} > \text{IORad}$: the within-family genetic contribution was underestimated.

Verification

$$\text{IORad}_{\text{RC}} - \text{IOLib}_{\text{RC}} = \rho_{\text{between}}^2 \Delta_{\text{between}} + \rho_{\text{within}}^2 \Delta_{\text{within}}$$

Methodological notes

Consider both h^2 scenarios The two sources (Becker vs. Howe) differ both in population and in method (GREML on WLS vs. within-family design in cohorts including MoBa). For WLS, Becker’s estimate is sample-specific but has a large SE (0.103) and requires an assumption to obtain h_{within}^2 . Howe’s estimate is more precisely measured for the within-family component but comes from a different population.

Uncertainty in estimation. In the current method the correction factors ρ_{between} and ρ_{within} are held fixed at their full-sample values across iterations. The variance components (and hence Δ_{between} , Δ_{within} , w , Sibcorr) vary across bootstrap samples. R^2_{between} and R^2_{within} are estimated from the full sample and held fixed.

Considerations: First, bootstrap SEs capture sampling variability in the variance components but not uncertainty in the external h^2 estimates. How to consider this too? Second, R^2 could be recomputed on each bootstrap resample. In practice the impact is small: R^2_{between} is estimated from N_{families} observations and is more stable than the lmer variance components that drive most bootstrap variance. In addition, having a single correction factor for reporting is clearer conceptually.

h_{within}^2 scaling for Becker is an assumption. The proportional scaling $h_{\text{within,Becker}}^2 = 0.04 \times (0.172/0.13)$ assumes the within/total heritability ratio is the same in WLS as in MoBa. The alternative — using Howe’s $h_{\text{within}}^2 = 0.04$ unchanged with Becker’s $h_{\text{pop}}^2 = 0.172$ — would be internally inconsistent, artificially inflating h_{between}^2 .

References

- A) Becker, Joel, et al. “Resource profile and user guide of the Polygenic Index Repository.” *Nature human behaviour* 5.12 (2021): 1744-1758.
- B) Howe, Laurence J., et al. “Within-sibship genome-wide association analyses decrease bias in estimates of direct genetic effects.” *Nature genetics* 54.5 (2022): 581-592.